



NETRA KULKARNI

Data Scientist | Business Scientist | Natural Language Processing | Deep Learning | AI-ML | Project Manager | Business Analyst - Retail | Technical Consultant

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EDUCATION

MS in Data Science & Engg

Instituto Superior Tecnico -
Universidade de Lisboa

08/2021 - Present Lisbon, Portugal

B.E.(Hons.) Computer Science

Rajiv Gandhi Technical
University

07/2004 - 06/2008 Indore, India

STRENGTHS

ML - DL - NLP

Built a Top 'n' similarity search algorithm employing natural language processing (NLP) and ANN (Approximate nearest neighbor) for live data

Ongoing Thesis on Machine Learning based Ontology matching harnessing the power of machine learning and natural language processing to elevate ontology matching, enabling precise semantic alignment and knowledge integration.

Project Management

Led a team of 3+ through an end-to-end project that enabled a brand new time-series based forecasting functionality for a leading retail chain (Siam Makro Thailand)

Design Thinking

Initiated efficiency optimisation discussions with top management; Identified gaps, proposed and built tools that streamlined processes leading onto an improved customer service plus saved over 40 man-hours per week across all project teams in the company.

Effective Leadership

Initiated and brought together other project managers to build a common knowledge repository for the event-support processes that improved the service quality while reduced the training time by almost 50%.

SUMMARY

One of the constants, in this fast changing world, is continuity.

With 12+ years of technology and ERPs under my belt, I rapidly transitioned into project and team management and now with an MS in Data Science, I seek continuity and ought to leverage my rich experience into my **Data Scientist** role.

Powered by new skills and knowledge in the **Data Science** Technologies, honing my expertise in Python (including Pandas, Scikit-learn, Numpy, PyTorch), machine learning, deep learning, information visualization, multivariate analysis, decision support models, and my most significant skill development in Natural Language Processing (NLP) from a world-class institute, Instituto Superior Tecnico - Universidade de Lisboa, I now look for the opportunities to innovate world-class solutions to the critical challenges faced by corporations and the global population.

I value building lasting connections and collaborating with fellow unconventional thinkers.

PROJECTS

ML Based Ontology Matching

03/2022 - Present

Lisbon Semantic Data Lab

As a data scientist, I researched how Large Language Models (LLMs) like S-BERT can revolutionize ontology matching, addressing challenges in semantic understanding and context.

- Challenge: Overcoming limitations in traditional ontology matching methods.
- Tasks: Developed innovative techniques for fine-tuning Large Language Models on ontology datasets.
- Generated: Enhanced ontology alignment accuracy by leveraging contextual information and LLM capabilities.
- Benefit: Facilitated more accurate data interoperability and knowledge sharing within complex domains.

Top 'n' Similarity Recommendation

10/2022 - 02/2023

Confidential Company

As a Data Scientist, I analysed large database, researched, feature engineered and modelled an algorithm that achieved the target KPIs

- Challenge: To develop a similarity search algorithm that lists top n matches for any given record from a large scale relational database (800K+ records)
- Tasks: data collection, cleaning, EDA, text processing, feature engineering, indexing with ANNOY
- Generated: a production-ready algorithm with acceptable results and target performance values
- Benefit: multi-language support + scalable processing

Hand Gesture recognition

05/2022 - 06/2022

Learning Based Multimedia Processing Course Project

As a Data Scientist, I modelled an algorithm that can identify hand gestures through an ordinary camera

- Challenge: To identify and classify 15 hand gestures read through an ordinary camera (sensor)
- Tasks: Researched a dataset with 14 gestures, added one of our own gesture Using Google's open-source framework *Mediapipe* to transform the 3D skeleton image, employed transfer learning on a pre-trained multi-channel deep convolution neural network (MC-DCNN)
- Generated: a model that could identify and classify hand gestures
- Benefit: ease to add new gestures, faster training with transfer learning

STRENGTHS

Empathy

Have been a part of several multi-cultural, multi-national and multi-nationality teams across multiple continents, and thus had the privilege to experience diversity that help empathize and understand colleagues and clients better.

Multi-Lingual

English (Adv.), Hindi (Native), Marathi (Native), Thai (Beg.), Portuguese (Beg.)

SKILLS

Machine Learning · Deep Learning ·

Info Viz · Multivariate Analysis ·

Decision Support Models · Cloud AWS ·

Product development & services ·

Engineering Economics ·

Business Strategy · Design Thinking ·

Smartsheet · Asana

Programming

Python · MySQL · R · C/C++ · Unix ·

Shell Scripting · Oracle SQL PL/SQL · D3 ·

OBIEE · RDBMS · PyTorch

Industry

Learning Management · EdTech ·

Retail Merchandising ·

Retail Demand Forecasting ·

Manufacturing IT · Logistics

MY PASSIONS

Recreational Blogging

Travelling

Leading a Cultural Community through events and social gatherings

Public Speaking

PROJECTS

Binary Classification for multi-variate data

12/2021 - 02/2022

Multivariate Analysis Course Project

As a Data Scientist, I modelled a Binary classification based on the customer details.

- Challenge: To predict if the client would subscribe to a bank term deposit based on multiple variables (20+)
- Tasks: EDA, dimensionality reduction using PCA, supervised learning and accuracy comparisons between multiple models; clustering to identify customer clusters in the data
- Generated: comparisons for multiple different model performances (accuracy %)
- Benefit: visible clusters in the customer data

Oracle Retail Demand Forecasting

02/2017 - 05/2018

Siam Makro PCL Thailand, (Cash and Carry)

As a Project Manager, led the implementation of time-series based retail demand forecasting module (ORDF) - part of the planning stack

- Challenge: Sprints of Technical implementation and subsequent analysis and tweaks for accurate predictions; Introduced sophisticated demand forecasting to the business team
- Tasks: Import historical sales data, clean and label it, analyse the predictions and tweak parameters to forecast better, add seasonality and causal info
- Generated: Demand forecasts for normal, seasonal and causal sales
- Benefit: Auto-replenishment for 'convenience' goods (about 10%)

PROFESSIONAL EXPERIENCE

Project Manager

11/2020 - 08/2021

MULTILEARNING Group

Lisbon, PORTUGAL

Edtech

Project Management | Project Co-ordination | Business Analysis | Process Streamlining | Virtual Events and E-learning

Business Analyst (Manager)

10/2013 - 05/2018

Siam Makro Public Company Ltd

Bangkok, THAILAND

Retail

Project Management | Business Analysis | Change management | Vendor Evaluation & Selection | Reporting to Key Stakeholders | Whitepapers | End - User Training

Oracle Apps Developer

04/2013 - 09/2013

Food and Agriculture Organization of the United Nations

Bangkok, THAILAND

Non-Profit

Software Development | Global Support | Social Initiatives

Technical Consultant

02/2012 - 03/2013

OLR

Bangkok, THAILAND

Retail IT

Requirement Gathering and Designing | Application Development | Application Support | Training

Software Developer

03/2011 - 12/2011

NIIT Technologies

Bangkok, THAILAND

Logistics IT

Requirement Gathering & Designing | Application Development & Support | Training

Software Engineer

10/2008 - 10/2010

KPIT

Pune, INDIA

Manufacturing IT

Trainee | Offshore Development | Global Support | System Integration